One Checkpoint for Complete, Slide-Only and RNA-Only Patients: Auditing and Training Histology–Transcriptomics Survival Models for Missing Inputs

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***Abstract*—Survival models that fuse whole-slide images with transcriptomics are evaluated on patients who have both inputs, whereas a hospital cohort also holds slide-only and RNA-only patients. Two questions are then open: does the fused model use both inputs, and what happens when one is missing? We audit this on five cohorts of The Cancer Genome Atlas under one equal-budget, three-seed protocol by re-evaluating trained checkpoints with an input removed or permuted across patients. Under this protocol, the official SurvPath implementation ranks patients the same way when its RNA input is permuted (C-index change $+0.001$) but loses 0.068 when the slide is permuted and falls to 0.47–0.52 without it. A query-bottleneck fusion model trained with learned null codes and modality dropout is equally invariant to RNA removal but loses only 0.030 without the slide. Adding auxiliary unimodal heads makes the fused prediction respond to RNA (-0.023 under permutation); the resulting single trained model (one checkpoint) keeps a C-index of 0.54–0.62 with either input removed and moves at most 0.025 when half the patients lack a modality, so one model serves complete, slide-only and RNA-only patients. With both inputs it does not detectably beat an RNA-only multilayer perceptron or a late-fusion ensemble, and its $+0.036$ margin over the untuned SurvPath is significant under the Wilcoxon test but not the t-test. Replayed on stored training histories, validation-loss early stopping would have cost the two variants 0.06–0.09 C-index and SurvPath nothing. Code, per-run results and checkpoints are released.**

***Index Terms*—Computational pathology, missing modality, model auditing, multimodal fusion, survival analysis, transcriptomics, whole-slide images.**

I. Introduction

In routine care, tumour sequencing is ordered for a subset of patients and slide digitisation is incomplete, so a prognostic cohort is rarely complete: some patients have a whole-slide image (WSI), some have bulk RNA-seq and some have both. Prognostic models that fuse the phenotype recorded by histology with the molecular state recorded by transcriptomics report higher concordance than either modality alone on most cohorts of The Cancer Genome Atlas (TCGA) [1]–[3]: pathology foundation models [4]–[6] turn a gigapixel WSI into a few thousand frozen patch embeddings, and co-attention between patch tokens and gene or pathway tokens [2], [7] relates the two sets. Such models are evaluated on patients who have both inputs, which leaves two questions open that an accuracy table cannot answer. Does the fused model use both inputs? A concordance index (C-index) measures how well predicted risks rank patients; it cannot show which input produced the ranking, so a model that silently ignores one input is a hidden failure. And what happens when one input is missing? A model that needs both inputs cannot serve the incomplete patients.

We answer both questions with an input-use audit: every trained checkpoint is re-evaluated with one input removed, with one input permuted across patients, and with a random fraction of patients lacking a modality, and its fused risk scores are correlated with the risks of its own single-input evaluations. Applied under one protocol to the official SurvPath implementation [2], the audit finds that SurvPath's ranking does not respond to its RNA input (ΔC-index $+0.001$ under permutation across patients) and collapses when the slide is permuted (ΔC-index -0.068) or removed (C-index 0.47–0.52).

The practical result is a training recipe that lets one trained model serve complete, slide-only and RNA-only patients. We train one query-bottleneck fusion model, PathQ-Former, in two variants that differ only in training. With learned null codes and modality dropout alone, the model is as invariant to RNA as SurvPath (ΔC-index 0.000 with RNA removed) but no longer collapses without the slide (-0.030 instead of SurvPath's -0.073). Adding auxiliary unimodal survival heads makes the fused prediction respond to RNA (-0.023 under permutation); the resulting checkpoint keeps a C-index of 0.54–0.62 with either input removed and changes by at most 0.025 when half of the patients lack a modality. With both inputs present, neither variant is detectably better than an RNA-only multilayer perceptron (MLP) or a late-fusion ensemble of two unimodal models, so the recipe buys coverage of incomplete patients, not accuracy. Our contributions are:

- A one-checkpoint training recipe assembled from existing components (null codes, modality dropout, auxiliary unimodal heads), with matched-variant evidence on where its missing-modality robustness comes from (Sections IV-C and IV-D).
- An input-use audit (removal, permutation across patients, risk-score correlation; 3 seeds $\times$ 5 folds $\times$ 5 cohorts) applied to a published co-attention model under an equal-budget protocol (Section IV-A). SurvPath's ranking, as

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trained under this protocol, does not respond to its RNA input; a high fused C-index does not show that both modalities are used.
- Protocol evidence that this literature rarely reports (Sections IV-B and IV-E): equal-budget accuracy with seed-aware statistics (seed-averaged 25-fold paired tests with bootstrap confidence intervals and Holm correction), and a post-hoc replay of checkpoint selection on event-poor folds from the stored training histories.

Scope of claims. SurvPath was run with its published hyper-parameters and was not re-tuned under our protocol. PathQ-Former's hyper-parameters were developed on the bladder cohort (BLCA) under a 10-epoch budget, and the auxiliary-head variant and its loss weight were adopted after the 20-epoch results on all five cohorts had been seen, so both variants are reported as co-primary and no accuracy superiority over SurvPath is claimed. The training recipe has not been applied to SurvPath. All results are on TCGA with frozen UNI2-h features, simulated missingness is random and therefore uninformative, and no clinical covariates are modelled.

II. Related Work

Fusion of WSIs and genomics. WSIs are processed as bags of patches under multiple-instance learning: attention-based MIL (ABMIL) [8], CLAM [9] and TransMIL [10] pool patch features from self-supervised [11] or foundation-model encoders trained on 10^8 patches or more [4]–[6]. Early fusion work concatenated convolutional neural network (CNN) and genomic features [12] or combined unimodal representations with Kronecker products and gating [13]; a pan-cancer study found that fusion improved concordance in 12 of 14 cancer types [1]. MCAT [7] introduced co-attention from functional gene groups to patches; SurvPath [2] replaced the groups with Reactome and Hallmark pathway tokens and modelled dense pathway–patch interactions in one Transformer; MOTCat [14], CMTA [15], PIBD [16] and MMP [3], which reports gains over SurvPath, vary the interaction, and continually evolved multimodal foundation models extend the setting [17]. Two-bank learned-query fusion over patches and pathways already exists in APL [18]; the query bottleneck used here is not claimed as new.

Missing modalities in multimodal survival. Modality dropout was introduced for gesture recognition [19] and applied to pan-cancer prognosis by Cheerla and Gevaert [20] and MultiSurv [21]. Ma et al. [22] showed that multimodal Transformers are not robust to missing inputs by default; SMIL [23], ShaSpec [24] and missing-aware prompts [25] address it with meta-learning, shared–specific features or prompt tokens. For WSIs and genomics, G-HANet [26] distils genomic knowledge into a histology-only model, and recent work binds incomplete modalities [27], distils prompts [28], learns conditional latent variables [29], uses prototype-guided cross-modal knowledge [30], memory-augmented hypergraphs [31] or prototype-driven pretraining [32]. None of these audits whether a published co-attention model uses its genomic input, nor tests under equal budgets whether unimodal auxiliary supervision changes that; this is the gap we address.

TABLE I
COHORTS AS REALISED BY THE DSS SPLITS (EVERY METHOD AND SEED USES THE SAME FIVE FOLDS). PATIENTS AND EVENTS PER VALIDATION FOLD ARE THE MINIMUM–MAXIMUM OVER THE FIVE FOLDS. †DEVELOPMENT COHORT FOR PATHQ-FORMER'S HYPER-PARAMETERS.

Cohort	Site	Patients	DSS events	Patients/fold	Events/fold
BLCA†	bladder	359	113	70–75	16–25
BRCA	breast	868	60	166–182	7–20
COADREAD	colorectal	296	37	45–71	5–13
HNSC	head & neck	392	117	56–126	13–45
STAD	stomach	318	84	42–101	9–33

Learned queries and modality imbalance. A set of learned vectors that cross-attend to a large input and return a fixed-size summary appears as inducing points [33], object queries [34], the Perceiver latent array [35], the Perceiver Resampler [36], the Q-Former [37] and the prototype banks of APL [18]; the property used here is that the output has a fixed shape, so an absent input can be replaced by a learned tensor of the same shape. Gradient Blending [38] and OGM-GE [39] showed that joint training lets one modality dominate and counteract it with per-modality supervision or gradient modulation. The auxiliary unimodal heads used here are unimodal supervision in that tradition, not a new idea; our contribution is to measure what they change in a survival-fusion model under an audit.

III. Materials and Methods

A. Data and Cohorts

Five TCGA cohorts are used (Table I): bladder urothelial carcinoma (BLCA), breast invasive carcinoma (BRCA), colorectal adenocarcinoma (COADREAD), head and neck squamous cell carcinoma (HNSC) and stomach adenocarcinoma (STAD). Patch embeddings are the released 1,536-d UNI2-h features [4], [40], [41], used as released. The RNA-seq matrices (4,999 genes), clinical metadata and five-fold patient-level splits are those released with SurvPath [2]. The primary endpoint is disease-specific survival (DSS) from the TCGA Clinical Data Resource [42]; overall survival (OS) is secondary. The split is fixed per cohort, so fold k holds the same validation patients for every method and seed, and the number of events in a validation fold ranges from 5 to 45 (median 16).

B. Protocol

One evaluation protocol is applied without change to every method (Supplementary Table S1). The unit of analysis is the patient, with all slides of a patient concatenated. Every method trains for a fixed budget of 20 epochs and reports the final checkpoint, the convention of MCAT and SurvPath; no checkpoint is selected on the validation fold, which is also the reported fold. The budget was set after inspecting pooled validation curves under a 10-epoch budget (Supplementary Section S-II), so 10-epoch results are reported as well. Time bins are the quartiles of uncensored training patients, with outer edges $\pm\infty$, reused for validation. The primary metric is Harrell's C-index on continuous time [43], [44]; the secondary metrics are Uno's inverse-probability-of-censoring-weighted

(IPCW) C-index [45] truncated at the 75th percentile of training event times, the integrated Brier score [46] on the interior quartile grid, and the log-rank test at the median risk split (Supplementary Sections S-III and S-VI). Three seeds (0, 1, 2) are run, with the fold index added to the seed. Supplementary Section S-VII lists evaluation pitfalls of survival MIL pipelines that the protocol avoids.

Statistics. The three seeds of a fold share its validation patients, so (fold, seed) pairs are not independent replicates [47]. The unit of analysis is therefore the (cohort, fold): the C-index of each method is averaged over the seeds available for both methods, the paired difference is formed on each of the 25 folds, and a paired t-test, a two-sided exact Wilcoxon signed-rank test and a 95% percentile-bootstrap confidence interval (CI) of the mean difference (10,000 resamples of folds) are computed over the folds. Per-cohort tests have five folds each; their p-values are Holm-adjusted over the five cohorts and are exploratory. "Mean $\pm$ sd" in a table is the mean and standard deviation over seeds of the five-fold mean unless the caption says otherwise.

C. Methods

Supplementary Table S2 lists the methods and the seeds run for each. SurvPath (official implementation, 21.2 M parameters), ABMIL [8], a self-normalizing network (SNN) [48] on RNA and a two-layer RNA MLP use the hyper-parameters of the SurvPath repository; none was tuned under this protocol. SurvPath samples 4,096 patches per slide, its published setting; PathQ-Former reads all patches of a patient (6,214 for the median of the 40 profiled BLCA patients, Supplementary Table S14). SurvPath's C-index values here are those of its official implementation under this protocol and with these patch features, not a reproduction of the values reported by its authors; every comparison in this paper is within the protocol of Section III-B, and the audit describes the checkpoints trained here, not the published model. Two further baselines are a slide-only and an RNA-only PathQ-Former, trained with the other modality absent, and a late-fusion ensemble that averages the z-scored risks of the two (same seed; two seeds). The two co-primary models are PathQ-Former without auxiliary heads (null codes and modality dropout; 16.9 M parameters) and PathQ-Former (the same plus auxiliary heads, $\lambda = 0.5$), trained with AdamW [49] (SurvPath: RAdam [50], its published setting; Supplementary Table S3); each is run with three seeds.

PathQ-Former. Fig. 1 shows the model; equations and hyper-parameters are in Supplementary Section S-II, compute cost in Supplementary Table S14 and the attention over pathway tokens, which is descriptive only [51], in Supplementary Fig. S2. Patient i has patch embeddings $X_i^h \in \mathbb{R}^{N_i \times 1536}$ from all of the patient's slides and a gene vector $x_i^g \in \mathbb{R}^{4999}$, min–max scaled to $[-1, 1]$ on the training split. Genes are grouped into $P = 275$ pathway tokens by per-pathway two-layer MLPs, using the Reactome [52] and MSigDB Hallmark [53] definitions released with SurvPath (331 pathways) with our filter of 3–300 member genes. Each modality $m \in \{h, g\}$ has $K = 32$ learned queries and two pre-norm blocks of cross-attention to its input tokens, self-attention among the queries and a feed-forward layer, which return a code $Z^m \in \mathbb{R}^{32\times256}$. The two codes are concatenated and passed through two Transformer encoder layers [54]; attention pooling and a sigmoid layer give $T = 4$ discrete-time hazards h_t with bins at the quartiles of the uncensored training times, survival $S(t) = \prod_{k\le t}(1 - h_k)$ and risk $r = -\sum_t S(t)$, as in MCAT and SurvPath [55]. For a patient with bin y and censoring indicator c the loss is the negative log-likelihood

$$\mathcal{L}(y, c) = -(1 - c)\big[\log S(y - 1) + \log h_y\big] - c \log S(y), \quad (1)$$

with the censored term unweighted for PathQ-Former and weighted by 0.5 for SurvPath, its published setting. *Training for missingness.* A learned null code $N^m \in \mathbb{R}^{32\times256}$ replaces Z^m when modality m is absent. During training each present modality is dropped with probability 0.15 per sample, independently, with at least one modality kept, and the dropped code is replaced by its null code, so the fusion Transformer sees null codes in training in the form it sees them at test time. *Auxiliary heads.* In the full model, attention pooling of Z^h and of Z^g each feeds a linear hazard head trained with (1) on the samples in which that modality is present; the objective is $\mathcal{L}_{\text{fused}} + \lambda \sum_m \mathbf{1}[m \text{ present}]\, \mathcal{L}_m$ with $\lambda = 0.5$, and the heads are discarded at test time. "PathQ-Former without aux. heads" is the same model trained with $\lambda = 0$.

D. The Input-Use Audit

Each trained checkpoint is evaluated on its validation fold under five conditions: *Complete*; *RNA removed*; *RNA permuted*, where the gene vectors are shuffled across the validation patients so that every patient receives a real but wrong profile; *Slide removed*; and *Slide permuted*. For PathQ-Former a removed modality is replaced by its null code. SurvPath cannot run without both inputs, so it receives the training-mean gene vector or the training-mean patch embedding in place of a removed one; PathQ-Former is also evaluated under the same mean imputation, to separate the effect of the null code from that of the architecture. Permutation is the stronger test: a mean or null input is neutral, whereas a permuted input is plausible and wrong, so a model that reads it must lose concordance. In the *random-missingness sweep*, 10, 20, 30 or 50% of validation patients lack one modality at random (five draws per fold, mean reported). Finally, the *risk-score correlation* is the Spearman ρ between a checkpoint's fused risks and the risks of its own RNA-removed and slide-removed evaluations on the same fold, and between its fused risks and those of the other methods on the same fold and seed. The variant without auxiliary heads has no permuted evaluation.

IV. Results

A. Do the Models Use Both Inputs?

Table II is the audit and Fig. 2 plots its pooled differences. SurvPath's C-index is unchanged when its RNA input is removed or permuted across patients ($+0.001$ in both cases; Table II gives every p-value): the model ranks patients the same way with their own profile, the training mean or another patient's profile. Removing the slide costs it 0.073

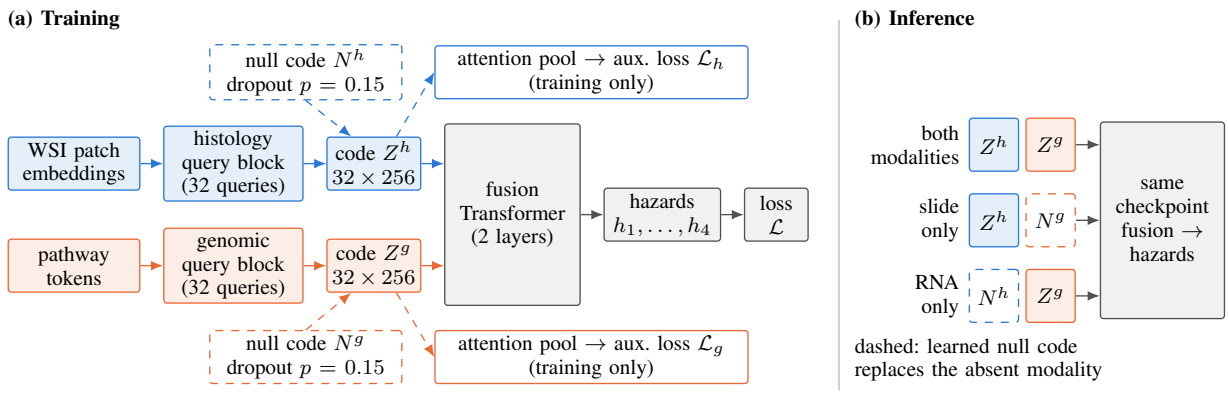


Fig. 1. PathQ-Former is trained for missingness and served from one checkpoint. (a) Training. The WSI patch embeddings and the pathway tokens are each read by a block of 32 learned queries, giving codes $Z^h, Z^g \in \mathbb{R}^{32\times256}$ that a two-layer fusion Transformer maps to four discrete-time hazards and the survival loss $\mathcal{L}$. With probability $p = 0.15$ per sample, modality dropout replaces a present code by its learned null code N^h or N^g (dashed), and auxiliary survival losses $\mathcal{L}_h$ and $\mathcal{L}_g$ on the attention-pooled codes make each branch prognostic on its own (training only). (b) Inference. Patients with both modalities, a slide only or RNA only enter the same checkpoint; the code of the absent modality is replaced by its null code and nothing else changes.

($p = 0.003$ and 0.002) and permuting it 0.068, leaving 0.47–0.52 (removed) and 0.46–0.53 (permuted) on every cohort; the smallest change under removal is on BRCA (-0.016; -0.004 under permutation), where its complete-input score (0.536) is already close to chance. PathQ-Former without auxiliary heads behaves the same way with respect to RNA (RNA removed $+0.000$) but loses only 0.030 without the slide. With auxiliary heads the fused prediction responds to RNA: -0.013 when it is removed and -0.023 when it is permuted ($p = 0.075$ and 0.063 over the 25 seed-averaged folds, short of 0.05; the attribution rests on this direction and on the contrast with the $+0.000$ and $+0.001$ of the other two models), the ordering expected of an input that is read; removing the slide costs 0.025 and permuting it 0.067 ($p = 0.0002$ and 0.0004), and the C-index stays at 0.54–0.62 with either input removed.

The risk-score correlations say how the fused prediction is composed. For PathQ-Former, the Spearman ρ between the fused risk and the same checkpoint's RNA-removed risk is 0.83 pooled over folds and seeds, against 0.45 with its slide-removed risk, and the two single-input risks correlate at 0.15 (Supplementary Table S9). The fused prediction therefore remains histology-led: the auxiliary heads make it respond to RNA, they do not make the two inputs symmetric. Its correlations with the other models' risks are in Supplementary Table S9.

B. Accuracy at Equal Budget

Table III gives the C-index of every method and Table IV the seed-averaged paired tests (Supplementary Table S4 adds the unimodal baselines). Pooled over 25 folds, PathQ-Former exceeds SurvPath by 0.036 (95% CI $[-0.004, +0.071]$; t-test $p = 0.074$, Wilcoxon $p = 0.030$; 18 of 25 folds) and the variant without auxiliary heads by 0.037 ($[+0.009, +0.065]$; $p = 0.020$ and 0.020; 17 of 25). Per cohort, BRCA is the only cohort with a raw $p < 0.05$ ($+0.086$, t-test $p = 0.017$), and it does not survive Holm correction over the five cohorts ($p = 0.084$); no DSS cohort survives correction (Supplementary Table S5). Against the RNA-only MLP the margins are $+0.012$ and $+0.014$ ($p \geq 0.34$), and against the late-fusion ensemble $+0.011$ and $+0.009$ ($p \geq 0.28$; seeds 0–1); the MLP scores higher than PathQ-Former on BRCA and COADREAD and higher than the variant without auxiliary heads on BLCA and BRCA. No unimodal baseline differs from SurvPath pooled (ABMIL $+0.001$, SNN -0.007, MLP $+0.024$). The equal-budget result is therefore parity with the best unimodal baseline, not superiority. On overall survival (one seed) PathQ-Former leads SurvPath by 0.034 ($[+0.014, +0.054]$; $p = 0.003$ and 0.003), and under a 10-epoch budget the variant without auxiliary heads leads by 0.028 ($p = 0.079$ and 0.063); both are tabulated in Supplementary Tables S6 and S7. Uno's IPCW C-index tracks Harrell's (Supplementary Table S8): pooled over the 25 seed-averaged folds, it exceeds SurvPath's by 0.039 for PathQ-Former and 0.046 for the variant without auxiliary heads (Wilcoxon $p = 0.042$ for both; t-test $p = 0.153$ and 0.045), and the integrated Brier score is lower than SurvPath's by 0.070 and 0.071 ($p < 0.001$; $n = 24$, one BRCA fold being undefined), as it is for ABMIL (-0.068) and the RNA MLP (-0.075). SurvPath's Brier score is the highest on every cohort (0.11–0.34): its published loss weights the censored term by 0.5 and leaves its hazards poorly calibrated under this protocol, so the Brier comparison is reported for completeness only.

C. Missing Inputs at Test Time

Fig. 3 shows the random-missingness sweep. With half of the validation patients missing one modality, the fused C-index of PathQ-Former changes by at most 0.025 (BRCA, 0.622 to 0.597) and that of the variant without auxiliary heads by at most 0.018 (BRCA). For PathQ-Former every change is within one sample seed standard deviation of the complete-input score (Table III); for the variant without auxiliary heads the HNSC and STAD changes (0.013 and 0.017) exceed it by 0.001 and 0.002. A site with mixed availability can instead route slide-only patients to ABMIL and RNA-only patients to the RNA MLP, or average the slide-only and RNA-only models when both inputs exist; routing and late fusion are robust by construction, and the late-fusion ensemble matches the joint model (Table IV). What the single checkpoint offers over routing is operational: one model to validate, version and monitor,

TABLE II

INPUT-USE AUDIT. C-INDEX OF THE SAME 20-EPOCH CHECKPOINT UNDER FIVE TEST-TIME CONDITIONS: MEAN ± SAMPLE STANDARD DEVIATION OVER THREE SEEDS OF THE FIVE-FOLD MEAN (AS IN TABLE III). A REMOVED INPUT IS THE TRAINING MEAN FOR SURVPATH AND THE LEARNED NULL CODE FOR PATHQ-FORMER; A PERMUTED INPUT IS SHUFFLED ACROSS THE VALIDATION PATIENTS. Δ: POOLED DIFFERENCE FROM COMPLETE OVER THE 25 SEED-AVERAGED FOLDS, WITH PAIRED t-TEST AND WILCOXON p-VALUES. THE VARIANT WITHOUT AUXILIARY HEADS HAS NO PERMUTED EVALUATION (N/A). †DEVELOPMENT COHORT.

Condition	BLCA†	BRCA	COADREAD	HNSC	STAD	Δ	t p	W p
SurvPath (official), mean imputation								
Complete	0.594 ± 0.012	0.536 ± 0.026	0.570 ± 0.038	0.552 ± 0.019	0.587 ± 0.008			
RNA removed	0.595 ± 0.013	0.536 ± 0.027	0.572 ± 0.037	0.553 ± 0.018	0.588 ± 0.007	$+0.001$	0.12	0.085
RNA permuted	0.594 ± 0.013	0.537 ± 0.028	0.572 ± 0.038	0.552 ± 0.011	0.587 ± 0.006	$+0.001$	0.36	0.13
Slide removed	0.486 ± 0.022	0.520 ± 0.025	0.473 ± 0.012	0.504 ± 0.043	0.493 ± 0.082	-0.073	0.003	0.002
Slide permuted	0.511 ± 0.018	0.532 ± 0.052	0.510 ± 0.024	0.489 ± 0.033	0.459 ± 0.016	-0.068	0.016	0.007
PathQ-Former without aux. heads, null codes								
Complete	0.609 ± 0.016	0.611 ± 0.039	0.638 ± 0.031	0.575 ± 0.012	0.593 ± 0.015			
RNA removed	0.612 ± 0.020	0.600 ± 0.046	0.633 ± 0.027	0.588 ± 0.008	0.595 ± 0.021	$+0.000$	0.96	0.51
RNA permuted	n/a	n/a	n/a	n/a	n/a			
Slide removed	0.590 ± 0.050	0.572 ± 0.033	0.601 ± 0.056	0.533 ± 0.017	0.578 ± 0.035	-0.030	0.080	0.16
Slide permuted	n/a	n/a	n/a	n/a	n/a			
PathQ-Former, null codes								
Complete	0.630 ± 0.010	0.622 ± 0.033	0.613 ± 0.023	0.582 ± 0.025	0.572 ± 0.022			
RNA removed	0.600 ± 0.002	0.592 ± 0.047	0.576 ± 0.025	0.596 ± 0.016	0.589 ± 0.010	-0.013	0.27	0.43
RNA permuted	0.597 ± 0.010	0.572 ± 0.046	0.618 ± 0.028	0.566 ± 0.019	0.555 ± 0.007	-0.023	0.075	0.063
Slide removed	0.613 ± 0.022	0.617 ± 0.052	0.583 ± 0.022	0.545 ± 0.026	0.538 ± 0.004	-0.025	0.19	0.11
Slide permuted	0.570 ± 0.035	0.539 ± 0.027	0.556 ± 0.033	0.539 ± 0.058	0.484 ± 0.035	-0.067	0.0002	0.0004

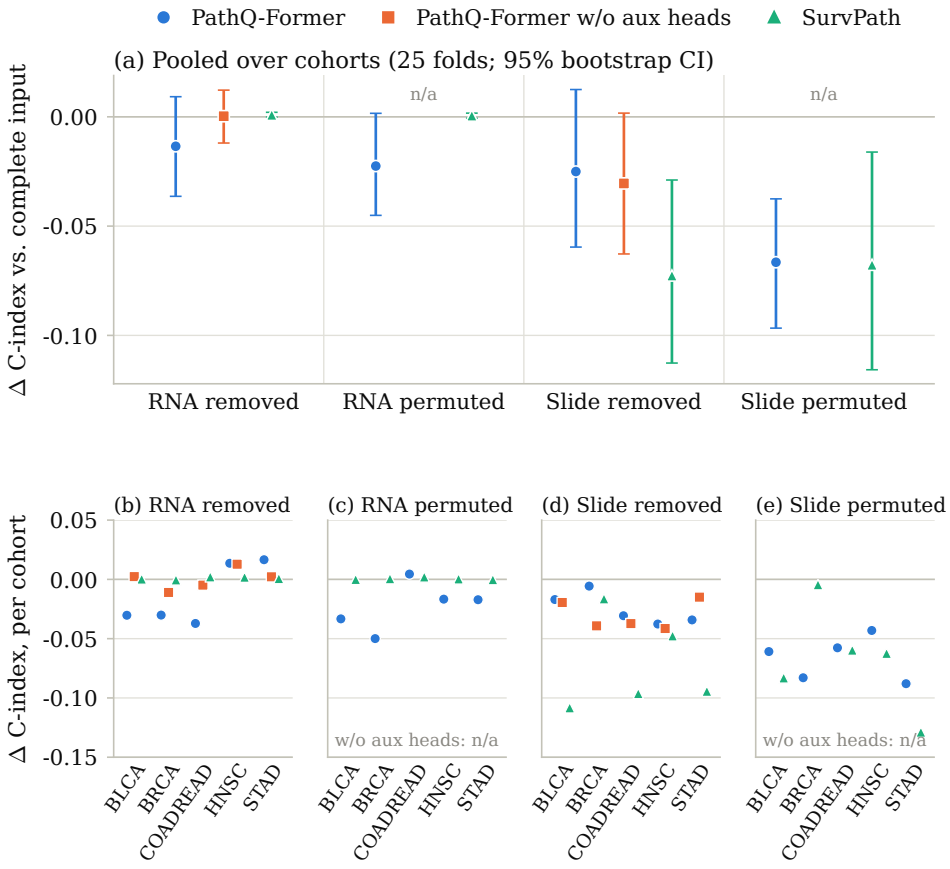


Fig. 2. Change in C-index of the same 20-epoch checkpoint when one test-time input is removed or permuted across validation patients, relative to its complete-input evaluation. In the legend, 'w/o aux heads' is the variant without auxiliary heads. SurvPath receives the training-mean gene vector or patch embedding for a removed input; PathQ-Former uses its learned null codes. (a) Pooled over the five cohorts: per (cohort, fold) the change is averaged over three seeds, then the mean and 95% bootstrap confidence interval over the 25 folds are shown (10,000 resamples of folds). (b)–(e) The same quantity per cohort (mean over five folds). The variant without auxiliary heads was not evaluated under permutation (n/a).

TABLE III
DISEASE-SPECIFIC SURVIVAL C-INDEX UNDER THE 20-EPOCH PROTOCOL: MEAN ± SAMPLE STANDARD DEVIATION OVER SEEDS OF THE FIVE-FOLD MEAN (THREE SEEDS; LATE FUSION TWO SEEDS; THE BRCA CELLS OF THE SINGLE-MODALITY PATHQ-FORMERS TWO SEEDS). POOLED IS THE MEAN OF THE FIVE COHORT MEANS. †DEVELOPMENT COHORT.

Method	Inputs	BLCA†	BRCA	COADREAD	HNSC	STAD	Pooled
PathQ-Former	WSI+RNA	0.630 ± 0.010	0.622 ± 0.033	0.613 ± 0.023	0.582 ± 0.025	0.572 ± 0.022	0.604
PathQ-Former without aux. heads	WSI+RNA	0.609 ± 0.016	0.611 ± 0.039	0.638 ± 0.031	0.575 ± 0.012	0.593 ± 0.015	0.605
SurvPath	WSI+RNA	0.594 ± 0.012	0.536 ± 0.026	0.570 ± 0.038	0.552 ± 0.019	0.587 ± 0.008	0.568
Late fusion	WSI+RNA	0.629 ± 0.018	0.606 ± 0.020	0.620 ± 0.069	0.571 ± 0.009	0.544 ± 0.006	0.594
ABMIL	WSI	0.566 ± 0.011	0.575 ± 0.004	0.587 ± 0.023	0.562 ± 0.028	0.553 ± 0.020	0.569
Slide-only PathQ-Former	WSI	0.592 ± 0.003	0.615 ± 0.027	0.574 ± 0.024	0.585 ± 0.003	0.582 ± 0.019	0.590
SNN	RNA	0.590 ± 0.009	0.558 ± 0.013	0.572 ± 0.028	0.535 ± 0.006	0.549 ± 0.009	0.561
RNA MLP	RNA	0.611 ± 0.012	0.627 ± 0.013	0.633 ± 0.016	0.555 ± 0.012	0.532 ± 0.009	0.592
RNA-only PathQ-Former	RNA	0.608 ± 0.020	0.559 ± 0.001	0.598 ± 0.035	0.554 ± 0.013	0.511 ± 0.005	0.566

TABLE IV
SEED-AVERAGED PAIRED TESTS OVER THE 25 POOLED (COHORT, FOLD) UNITS: MEAN DIFFERENCE IN C-INDEX (FIRST MINUS SECOND METHOD), 95% PERCENTILE-BOOTSTRAP CI (10,000 RESAMPLES OF FOLDS), PAIRED t-TEST AND EXACT WILCOXON p-VALUES, AND FOLDS WON. SEEDS ARE AVERAGED PER FOLD BEFORE PAIRING (THREE SEEDS FOR DSS; SEEDS 0–1 AGAINST LATE FUSION; ONE SEED FOR OS). THE UNIMODAL BASELINES AGAINST SURVPATH ARE IN SUPPLEMENTARY TABLE S4.

Comparison	Setting	Δ	95% CI	t p	W p	Wins
PathQ-Former vs SurvPath	DSS	$+0.036$	$[-0.004, +0.071]$	0.074	0.030	18/25
PathQ-Former without aux. heads vs SurvPath	DSS	$+0.037$	$[+0.009, +0.065]$	0.020	0.020	17/25
PathQ-Former vs RNA MLP	DSS	$+0.012$	$[-0.022, +0.044]$	0.47	0.34	14/25
PathQ-Former without aux. heads vs RNA MLP	DSS	$+0.014$	$[-0.023, +0.049]$	0.47	0.37	16/25
PathQ-Former vs late fusion	DSS, seeds 0–1	$+0.011$	$[-0.013, +0.034]$	0.39	0.28	14/25
PathQ-Former without aux. heads vs late fusion	DSS, seeds 0–1	$+0.009$	$[-0.020, +0.036]$	0.54	0.52	14/25
PathQ-Former vs SurvPath	OS, seed 0	$+0.034$	$[+0.014, +0.054]$	0.003	0.003	19/25
PathQ-Former without aux. heads vs SurvPath	DSS, 10 epochs	$+0.028$	$[-0.001, +0.057]$	0.079	0.063	18/25

and one output head whose risks are produced the same way for every patient; whether those risks are comparable across missingness patterns was not tested. The one-checkpoint result is about coverage, not accuracy: with the RNA removed, the single PathQ-Former checkpoint scores 0.576–0.600 against 0.553–0.587 for ABMIL (higher on four cohorts, lower on COADREAD); with the slide removed it scores 0.538–0.617 against 0.532–0.633 for the RNA MLP, higher on BLCA and STAD and lower on BRCA, COADREAD and HNSC (by up to 0.050 on COADREAD; three-seed means throughout).

D. Where the Robustness Comes From

Evaluating the same checkpoints under mean imputation instead of the null code separates the two. For the variant without auxiliary heads, replacing the removed slide by the training-mean patch embedding instead of the null code turns a loss of 0.030 into a loss of 0.069 (t-test $p = 3 \times 10^{-4}$, Wilcoxon $p = 2.5 \times 10^{-4}$; one seed, 25 folds), the size of SurvPath's collapse; with the heads the loss is 0.035 under mean imputation ($p = 0.040$ and 0.045; three seeds) against 0.025 with the null code (per-cohort values in Supplementary Table S10). The attribution is therefore: null codes and modality dropout remove the collapse without the slide; the auxiliary heads make the fused output respond to RNA; and the fused risk remains histology-led (ρ 0.83 against 0.45). Fusion-block ablations on BLCA under the 10-epoch budget (one seed) move the fused C-index within 0.595–0.638, inside the fold standard deviation of 0.07–0.13 (Supplementary Table S12). Under the 20-epoch budget on BLCA (one seed, trained on a CPU), capping PathQ-Former at the 4,096 training patches per slide that SurvPath uses gives 0.626 against 0.625 for the seed-0 run with all patches (0.630 ± 0.010 over three seeds), so on this cohort the margin over SurvPath is not explained by PathQ-Former reading more patches; two or eight time bins give 0.628 and 0.609, the 281 Xena pathway groups [56] 0.628 and the 50 Hallmark gene sets alone 0.600, all within one fold standard deviation (Supplementary Table S13).

E. Checkpoint Selection on Event-Poor Folds

Every run stores its per-epoch validation history, so the effect of a selection rule can be replayed without training. Supplementary Table S11 replays validation-loss early stopping with patience 5 on the 590 stored fold histories of the eight 20-epoch methods. The rule selects epoch 1 on median (mean 1.6; 93% of folds epoch 1–3) and would have stopped after 6.6 of 20 epochs on average, with the patience limit reached in every fold. Pooled over the 25 seed-averaged folds, the early-stopped checkpoint scores below the final one by 0.090 for PathQ-Former without heads, 0.059 for PathQ-Former, 0.065 for the slide-only and 0.034 for the RNA-only PathQ-Former, 0.038 for SNN and 0.027 for the RNA MLP, and by nothing for ABMIL (-0.001) and SurvPath ($+0.000$). The histories show why: ABMIL and SurvPath reach their best validation C-index at epoch 6.2 and 6.9 on average, SNN and the MLP at 7.2–7.3, and the PathQ-Former variants, which train with a warm-up epoch and a lower learning rate (Supplementary Table S3), at 9.3–10.0, so an epoch-1–2 checkpoint costs them most. Conversely, under the oracle rule (the epoch of highest validation C-index, an unattainable bound biased in favour of every method) the pooled C-index is 0.67–0.68 for SurvPath,

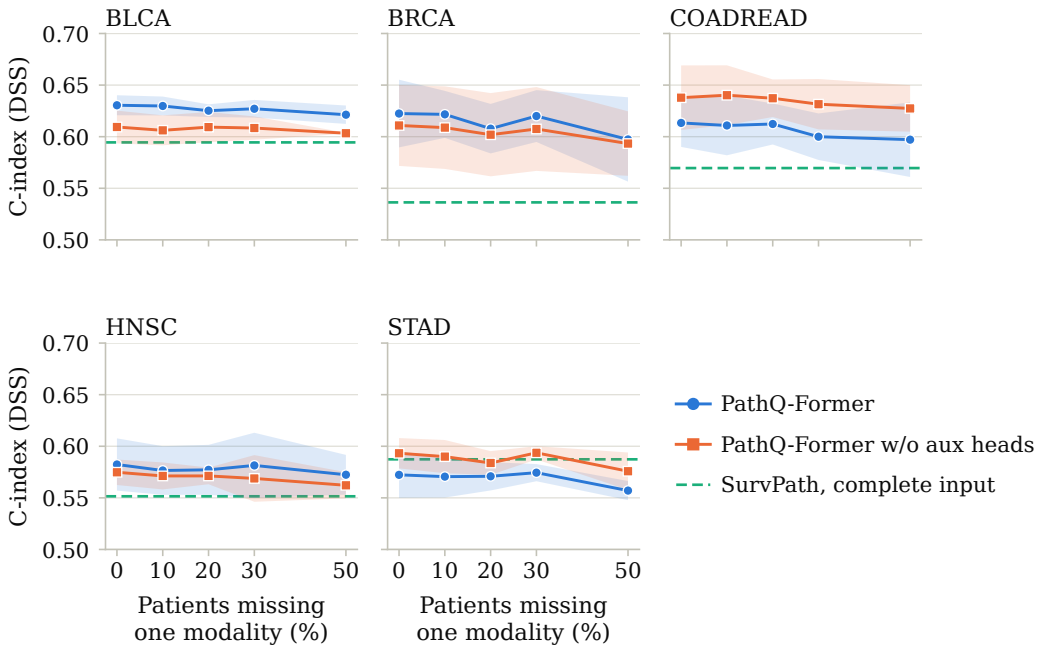


Fig. 3. C-index of PathQ-Former and of its variant without auxiliary heads ('w/o aux heads' in the legend; 20 epochs) when 0–50% of validation patients are missing one modality at random (mean of five random draws per fold). Lines: mean over three seeds of the five-fold mean; bands: ±1 sd over seeds. Dashed line: SurvPath with complete inputs (three-seed mean). With half the patients missing a modality the fused C-index changes by at most 0.025 (BRCA).

ABMIL, SNN and the RNA MLP and 0.69 for both PathQ-Former variants (Supplementary Table S11), so the margin in Table IV depends on the fixed-budget, final-checkpoint rule as well as on the models. Under the replayed rule 23 of the 40 method × cohort cells have a mean C-index within 0.45–0.55, against 5 under the final checkpoint (HNSC 8 of 8, STAD 7 of 8, BRCA 4 of 8). An earlier validation-loss campaign on these cohorts (patience 5, minimum 6 epochs; 145 folds) behaved the same way: median selected epoch 1 (82% of folds epoch 1–3) and 14 of 25 method × cohort cells within 0.45–0.55 (Supplementary Section S-VIII). Every replayed rule selects and scores on the same fold, so the measured deficit of early stopping is conservative. On cohorts with 5–45 events per validation fold, checkpoints should not be selected on the validation negative log-likelihood; a fixed budget with the final checkpoint, or at least the selected epochs and the events per fold, should be reported.

V. Discussion

A general lesson. Fusion of a strong and a weak modality can converge to ignoring the weak one with no accuracy signal: the fused C-index of the SurvPath checkpoints trained here is as high as their slide-only behaviour permits, and nothing in the survival objective penalises leaving the genomic branch unused once the fusion block has found a low-loss route through the histology branch. This is the modality-competition effect that Gradient Blending [38] and OGM-GE [39] describe for classification, and the same remedies apply: per-modality supervision supplies a direct gradient to the weak branch. The audit detects the failure without any training: removal and permutation across patients, with the risk-score correlation, show which input a ranking responds to, and they should accompany any multimodal C-index. Unimodal supervision makes the fused output respond to RNA, but the fused risk still correlates at 0.83 with the same checkpoint's RNA-removed risk and at 0.45 with its slide-removed risk, and the RNA-permuted loss (0.023) is a third of the slide-permuted loss (0.067).

The RNA-only baseline. A two-layer MLP on bulk expression, run with repository defaults, matches both fusion variants pooled and scores higher than PathQ-Former on two of five cohorts. This baseline is cheap and seldom reported at equal budget; without it, an improvement attributed to fusion may be an improvement attributable to the genomic branch. Separating "pathway tokens help" from "fusion helps" would need an MLP that receives the same pathway tokens, which we have not run.

Limitations.

- **Baseline tuning and variant selection.** SurvPath was not re-tuned, and PathQ-Former's hyper-parameters and the auxiliary-head variant were chosen as described under Scope of claims and in Supplementary Section S-II. The accuracy comparison is favourable to PathQ-Former by an amount that cannot be bounded without tuning the baseline, and its margin shrinks to about 0.01 under oracle selection (Section IV-E).
- **Recipe on other architectures.** Null codes, modality dropout and auxiliary heads have not been applied to SurvPath, so how much of the robustness is specific to the query bottleneck is untested.
- **Seeds and statistics.** Three seeds for the main comparison, two for late fusion, one for OS. Per-cohort tests have five folds; the fold-level standard deviation of the C-index is 0.12–0.25 on COADREAD for the fusion

models (0.05–0.25 over all methods), fold 3 of BLCA is weak for every method except the RNA-only PathQ-Former, and fold 0 of COADREAD, with five events, scores 0.85–0.92 for PathQ-Former across seeds. No per-cohort DSS difference survives correction; on OS (one seed) BLCA, the development cohort, does under the t-test only (Holm $p = 0.039$; Wilcoxon 0.312). The RNA-permuted loss of 0.023 that supports the auxiliary-head claim has $p = 0.075$ and 0.063 over 25 folds.

- **Data and clinical use.** TCGA only, with frozen UNI2-h features whose pretraining corpus may include TCGA slides, which would make every WSI-based result optimistic; no external cohort; missingness simulated at random, so informative missingness is not covered; no clinical covariates (stage, grade, age), so whether any model adds to them is untested; performance across sex, race and age subgroups was not evaluated. The model is a research prototype evaluated on public, de-identified data, intended for research-grade risk stratification of a cohort with mixed data availability from one model, not for patient-level decision support, and it must not be used to inform patient care.

VI. Conclusion

A fused histology–transcriptomics survival model can score well while its ranking ignores one input: the official SurvPath implementation, under an equal-budget protocol on five TCGA cohorts, does not respond to its RNA input and collapses without the slide. Removal and permutation audits with risk-score correlations detect this and should be reported with any multimodal C-index. Training one fusion model with null codes and modality dropout removes the collapse, and auxiliary unimodal heads make the fused prediction respond to RNA; the resulting single checkpoint serves complete, slide-only and RNA-only patients; with both inputs it is not detectably different from an RNA-only MLP or a late-fusion ensemble, and with one input absent it stays within 0.05 of the dedicated unimodal baseline. Replaying checkpoint selection on the stored histories shows that validation-loss early stopping would have selected epoch 1–3 checkpoints and penalised the late-learning models on these event-poor folds, which argues for fixed budgets and for reporting selected epochs and events per fold.

Data and Code Availability

TCGA data are public and de-identified. UNI2-h patch features [40], [41] and the RNA-seq matrices, clinical metadata and splits released with SurvPath [2] are available from their authors. The code (MIT licence), configuration files, archived per-run results (results, configuration, per-fold predictions and training histories for every run; inventory in Supplementary Section S-VIII) and the script that regenerates every table in Section IV and Supplementary Section S-III are public at https://github.com/SharoonSharif/PathQFormer and archived at Zenodo (https://doi.org/10.5281/zenodo.22900123). The 275 trained fold models (PathQ-Former and its variant without auxiliary heads at three seeds, SurvPath and the slide-only and RNA-only PathQ-Formers at seed 0, on the five cohorts, plus the two OS runs: 11 run configurations $\times$ five cohorts $\times$ five folds; 18 GB) are on the Hugging Face Hub at https://huggingface.co/sharoonsharif1/PathQFormer-checkpoints. The supplementary material (one PDF) contains the protocol and methods tables, the model equations and training configuration, the full tables, compute cost, pathway attention, Kaplan–Meier curves, evaluation pitfalls and the run inventory.

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